

COS 760 / Optimizing Cross-Lingual Embeddings for isiZulu, Sepedi, and Setswana: Alignment Strategies, Pivot Selection, and Morphological Challenges.

Subtitle: Alignment Performance and Data Efficiency Across Conjunctive and Disjunctive Bantu Languages

Group 40

Tom Balay, Ndamulelo Mulaudzi, Franco van der Westhuizen

1. Introduction

South Africa's linguistic landscape is incredibly diverse, yet languages like isiZulu, Sepedi, and Setswana remain severely underrepresented in NLP, limiting equitable digital accessibility. Cross-lingual word embeddings bridge this gap by projecting monolingual spaces into a shared multilingual space to enable transfer learning. However, alignment quality depends heavily on target language morphology. IsiZulu uses a conjunctive writing system (multiple morphemes form single tokens), whereas Sepedi and Setswana are largely disjunctive (morphemes as separate tokens). This proposal investigates (1) whether Canonical Correlation Analysis (CCA) outperforms VecMap and Kernel CCA (KCCA) across these distinct morphologies, and (2) how data volume requirements differ between conjunctive and disjunctive languages for downstream Named Entity Recognition (NER).

2. Background and Related Work

Cross-lingual alignment relies heavily on linear mapping approaches like VecMap (Artetxe et al., 2018) or correlation-based methods like CCA (Faruqui & Dyer, 2014). For Bantu languages, Modupe et al. (2023) demonstrated that CCA outperforms VecMap for Setswana and Sepedi, motivating a closer investigation into whether this holds for isiZulu's conjunctive orthography. To address non-linear structures, Kornblith et al. (2019) introduced Centered Kernel Alignment (CKA) to evaluate representational similarity, which we pair with KCCA to handle complex morphological divergences. Finally, regarding data efficiency, Eiselen & Gaustad (2023) showed conjunctive languages require significantly more data than disjunctive ones for POS tagging; we extend this to evaluate empirical learning curves for zero-shot NER transfer.

3. Proposed Methodology

Research Questions:

- **RQ1 (Alignment):** Does CCA outperform VecMap and KCCA when aligning isiZulu, Sepedi, and Setswana to English, and does morphological profile dictate the advantage of non-linear methods?
- **RQ2 (Data Efficiency):** How do data-volume requirements for acceptable NER F1 scores differ between conjunctive and disjunctive languages?

Data: We will utilize publicly available datasets: NCHLT Text Corpora (~1-2M tokens per language) for monolingual embeddings, MasakhaNER 2.0 (~25K sentences per language) for transfer learning, and Bilingual Seed Lexicons (~5K pairs) for alignment.

Approach:

1. **Training:** Train 100-D FastText embeddings (using subword n-grams to mitigate conjunctive OOV issues) on NCHLT and Wikipedia. For RQ2, we train on corpus subsets: 100%, 75%, 50%, 25%, 10%, and 5%.
2. **Alignment:** Implement VecMap (linear orthogonal), CCA (linear correlation), and KCCA (non-linear geometric) to align the Bantu languages to an English pivot.
3. **Transfer:** Train a BiLSTM-CRF NER model on English data (CONLL-2003) and evaluate zero-shot transfer to the target languages across all embedding subsets.
4. **Analysis:** Conduct qualitative error analysis on mistranslations and compute CKA scores.

Feasibility, Timeline & Work Plan: The project is highly feasible due to the use of public datasets. Computational overhead from KCCA will be mitigated by prototyping on small corpus subsets.

- *Weeks 1-2:* Data acquisition and FastText embedding training.
- *Weeks 3-4:* Cross-lingual alignment and zero-shot NER transfer.
- *Weeks 5-6:* Evaluation, geometric analysis, and report synthesis.
- **Tom Balay (Alignment Lead):** Implement, tune, and evaluate VecMap, CCA, and KCCA models.
- **Franco van der Westhuizen (Data/Transfer Lead):** Handle FastText training and build the BiLSTM-CRF NER pipeline.
- **Ndamulelo Mulaudzi (Evaluation Lead):** Compute P@k/CKA scores, plot learning curves, and conduct error analysis.

Table 1. The datasets to be used in the project.

Dataset	Languages Covered	Size
NCHLT Text Corpora	isiZulu, Sepedi, Setswana	~1M to 2M tokens per language
MasakhaNER 2.0 / SADIaR NER	isiZulu, Sepedi, Setswana	~25K sentences per language
Bilingual Seed Lexicons	isiZulu-English, Sepedi-English, Setswana-English	~5,000 pairs per language pair

4. Evaluation

RQ1 will be intrinsically evaluated using word translation accuracy (P@1, P@5) and mean cosine similarity on 1,000 held-out pairs. We will compute pre-alignment CKA scores to baseline geometric similarity. Extrinsic evaluation will report entity-level F1 scores on the MasakhaNER dataset. For RQ2, we will construct learning curves by plotting NER F1 scores

against log-scaled corpus sizes to find the break-even point (the minimum data required to surpass 50% F1), quantifying the efficiency gap.

5. Expected Output

We expect pre-alignment CKA scores to reveal that the isiZulu-English space is more geometrically divergent than Sepedi-English due to its conjunctive morphology. Consequently, KCCA should outperform linear methods specifically for isiZulu, while VecMap/CCA may suffice for Sepedi. For RQ2, we expect isiZulu's learning curve to be flatter, requiring substantially more data to reach the 50% F1 threshold. This will empirically confirm the conjunctive-language data disadvantage in a cross-lingual transfer setting.

6. References

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